

Research Profile

I am a Lecturer in Artificial Intelligence at the University of Bristol. My research develops **causal foundations for intelligent systems**. A central theme of my work is understanding how AI systems can move beyond correlation: learning meaningful structure, reasoning about causes and interventions, adapting when environments change.

Academic Appointments

University of Bristol, UK *Oct 2024 – Present*
School of Engineering Mathematics and Technology, MaVi Group
Lecturer (= U.S. Assistant Professor) in Artificial Intelligence
Research: causal AI, reinforcement learning, Human–AI interaction, foundation models, world models, and autonomous agents.

Education

University College London, UK *Sep 2020 – Aug 2025*
PhD in Computer Science
Supervisor: Prof. Jun Wang
Research: Causal Representation Learning, Reinforcement Learning, Multi-Agent Decision Making

University of Chinese Academy of Sciences, China *Sep 2017 – Jul 2020*
MSc in Computer Application Technology
Research: Causal Inference, Online Learning, Reinforcement Learning

Beijing Jiaotong University, China *Sep 2012 – Jul 2016*
Bachelor in Software Engineering

Selected Publications (Full list: [Google Scholar](#))

Causal Foundation Models and Agentic AI

- **ProcMEM: Learning Reusable Procedural Memory from Experience via Non-Parametric PPO for LLM Agents.** ICML 2026 **Spotlight**.
Qirui Mi, Zhijian Ma, **Mengyue Yang**, Yisen Wang, Haoxuan Li, Haifeng Zhang, Jun Wang.
- **A Principle of Pre-Strategy Intervention for Multi-Agent Reinforcement Learning.** NeurIPS 2025.
Anjie Liu, Jianhong Wang, Samuel Kaski, Jun Wang, **Mengyue Yang**.
- **Causal Sufficiency and Necessity Improves Chain-of-Thought Reasoning.** NeurIPS 2025.
Xiangning Yu, Zhuohan Wang, Linyi Yang, Haoxuan Li, Anjie Liu, Xiao Xue, Jun Wang, **Mengyue Yang**.
- **Curious Causality-Seeking Agents Learn Meta Causal World.** NeurIPS 2025.
Zhiyu Zhao, Haoxuan Li, Haifeng Zhang, Jun Wang, Francesco Faccio, Jürgen Schmidhuber, **Mengyue Yang**.
- **MF-LLM: Simulating Population Decision Dynamics via a Mean-Field Large Language Model Framework.** NeurIPS 2025.
Qirui Mi, **Mengyue Yang**, Xiangning Yu, Zhiyu Zhao, Cheng Deng, Bo An, Haifeng Zhang, Xu Chen, Jun Wang.
- **When Can Proxies Improve the Sample Complexity of Preference Learning?** ICML 2025.
Yuchen Zhu, Daniel Augusto de Souza, Zhengyan Shi, **Mengyue Yang**, Pasquale Minervini, Alexander D'Amour, Matt J. Kusner.
- **Rectifying Unfairness in Recommendation Feedback Loops.** SIGIR 2023.
Mengyue Yang, Jun Wang, Jean-Francois Ton.
- **Lending Interaction Wings to Recommender Systems with Plug-and-Play Conversational Agents.** NeurIPS

2023.

Jiarui Jin, Xianyu Chen, Fanghua Ye, **Mengyue Yang**, Yue Feng, Weinan Zhang, Yong Yu, Jun Wang.

Causal Representation Learning and Robust AI

- **Causal Fine-Tuning under Latent Confounded Shift.** ICML 2026.
Jialin Yu, Yuxiang Zhou, Haoxuan Li, Junchi Yu, **Mengyue Yang**, Yulan He, Nevin L. Zhang, Philip Torr, Ricardo Silva.
- **Causal Representation Learning from Multimodal Biological Observations.** ICLR 2025.
Xue Yan, Yan Song, Xidong Feng, **Mengyue Yang**, Haifeng Zhang, Haitham Bou Ammar, Jun Wang.
- **Invariant Learning via Probability of Sufficient and Necessary Causes.** NeurIPS 2023 **Spotlight**.
Mengyue Yang, Zhen Fang, Yonggang Zhang, Yali Du, Jean-Francois Ton, Jianhong Wang, Jun Wang.
- **Specify Robust Causal Representation from Mixed Observations.** KDD 2023.
Mengyue Yang, Xinyu Cai, Furui Liu, Xu Chen, Zhitang Chen, Jianye Hao, Jun Wang.
- **CausalVAE: Disentangled Representation Learning via Neural Structural Causal Models.** CVPR 2021.
Mengyue Yang, Furui Liu, Zhitang Chen, Jianye Hao, Jun Wang.

Foundation Models and Decision Making

- **Learning to Memorize with Attributive and Associative Memory for Online Test-Time Adaptation of Vision-Language Models.** ICML 2026.
Yuchao Zhang, Eric Wang, Fan Zhang, Haoxuan Li, Yisen Wang, Zhouchen Lin, Jun Wang, Qirui Mi, **Mengyue Yang**.
- **A Very Big Video Reasoning Suite.** ICML 2026.
Maijunxian Wang, ..., **Mengyue Yang**, ..., Hokin Deng.
- **MAD-Logic: Multi-Agent Debate Enhances Symbolic Translation and Reasoning.** ICLR 2026.
Haocheng Yang, Fengxiang Cheng, Tianjun Yao, **Mengyue Yang**, Jiajun Chai, Xiaohan Wang, Guojun Yin, Wei Lin, Soumya Kar, Fenrong Liu, Haoxuan Li, Yisen Wang.
- **Large Language Models are Demonstration Pre-Selectors for Themselves.** ICML 2025.
Jiarui Jin, Yuwei Wu, Xiaoting He, Haoxuan Li, Weinan Zhang, Yiming Yang, Yong Yu, Jun Wang, **Mengyue Yang**.
- **Efficient Reinforcement Learning with Large Language Model Priors.** ICLR 2025.
Yuewen Sun, Lingjing Kong, Guangyi Chen, Loka Li, Gongxu Luo, Zijian Li, Yixuan Zhang, Yujia Zheng, **Mengyue Yang**, Petar Stojanov, Eran Segal, Eric P. Xing, Kun Zhang.

Research and Industry Experience

KAUST , Visiting PhD Student Advised by Prof. Jürgen Schmidhuber . Research on causal world models.	<i>Aug 2024 – Sep 2024</i>
CMU & MBZUAI , Visiting PhD Student Advised by Prof. Kun Zhang . Research on causal representation learning.	<i>Mar 2024 – Jul 2024</i>
TikTok (ByteDance Research), London , Research Intern Causal representation learning and fairness. Supervised by Dr. Jean-Francois Ton and Dr. Hang Li.	<i>Feb 2022 – Jul 2022</i>
Microsoft STCA, Beijing , Software Engineer Intern Ads Data & AI Platform Team. Large-scale machine learning systems and Microsoft Azure.	<i>Jul 2019 – Oct 2019</i>
Didi AI Labs, Beijing , Research Intern Reinforcement Learning Team. Research on online learning and sequential decision-making.	<i>Sep 2018 – Jul 2019</i>

Teaching Experience

- **SEMT20007 Reinforcement Learning** — University of Bristol, **Unit Director and Lecturer**, 2026–2027, Undergraduate Year 2. Designed and lead a new unit covering reinforcement learning foundations, decision-making, experimentation, and independent technical reasoning; responsible for curriculum design, learning outcomes, teaching materials, and assessment.
- **SEMT30008 Advanced Methods in Artificial Intelligence** — University of Bristol, Lecturer, 2025–2026, Undergraduate Year 3. Teaching on modern AI and foundation models, including large language model foundations, pre-training, and post-training.
- **SEMT10005 Principles of Artificial Intelligence** — University of Bristol, Lecturer, 2025–2026, Undergraduate Year 1. Foundational AI teaching, including probabilistic modelling.

- **SEMTM0016 Artificial Intelligence for Robotics** — University of Bristol, Lecturer, 2024–2026, Postgraduate. Teaching reinforcement learning and decision-making for intelligent systems.
- **COMP0124 Multi-Agent Artificial Intelligence** — University College London, Teaching Assistant, 2021–2024, Postgraduate. Teaching support on strategic interaction, game-theoretic reasoning, coordination, and multi-agent learning.

Research Leadership & Community Building

- **Organizer and Program Chair**, ICLR 2026 Workshop: *The 2nd Workshop on World Models: Understanding, Modelling and Scaling*.
- **Lead Organizer and Program Chair**, NeurIPS 2026 Workshop on Physical Understanding for Decision-Making.
- **Co-organizer**, NeurIPS 2025 Workshop: *Embodied World Models*.
- **Organizer and Program Chair**, ICLR 2025 Workshop: *World Models: Understanding, Modelling and Scaling*.
- **Co-organizer**, AAAI 2025 Workshop: *Artificial Intelligence with Causal Techniques*.
- **Co-organizer**, NeurIPS 2024 Workshop: *Causality and Large Models*.
- **Co-organizer**, ICDM 2024 Workshop: *Causal Representation Learning*.
- **Co-organizer**, NeurIPS 2023 Competition: *Causal Structure Learning from Event Sequences and Prior Knowledge*.

Selected Invited Talks, Panels & Tutorials

- NeurIPS 2026 Workshop on World Models in Physical AI: Invited Keynote
- ICLR 2026 Workshop on Logical Reasoning for Large Language Models: Invited Keynote, *Grounded LLM Reasoning*.
- AAAI 2026 New Faculty Highlight: *Causal Foundation World Models*.
- ICLR 2026 Workshop on Multi-Agent Learning and Its Opportunities in the Era of Generative AI: Panel Discussion.
- AAAI 2026 Workshop on LLM-based Multi-Agent Systems: Panel Discussion.
- DAI 2025 Workshop on LLM-based Multi-Agent Systems: Invited Keynote and Panel Discussion.
- ICML 2025 Workshop on Multi-Agent Systems: Panel Discussion.
- ICML 2025 Workshop on Scaling Up Intervention Models: Invited Keynote, *Scaling up Causal Learning in Open-Ended Worlds*.
- IJTCS-FAW 2025: Invited Forum Talk, *From Games to Graphs: Causal Mechanisms for Steering Multi-Agent Outcomes*.
- DAI 2024 Tutorial: *Causality and Large Models*.
- ACML 2024 Tutorial: *Causality and Large Models*.
- ICDM 2024 Tutorial: *Causality and Large Models*.
- KAUST Rising Star in AI Symposium: *Causal Representation Learning*.

Academic Service

- **Area Chair**: NeurIPS 2026.
- Guest Editor, *Build Trust in Foundation Models: Interpretability, Safety, and Robustness*, Journal of Machine Learning Special Issue.
- Co-organizer, Causal AI Workshop with IHES, France.
- **Reviewer**: NeurIPS, ICML, ICLR, AAAI, AISTATS, KDD, CVPR, TPAMI, TNNLS, SDM.

Honours & Awards

- WWW workshop on CausalTFM Outstanding Paper Award
- **AAAI 2026 New Faculty Highlight**.
- **Rising Star in AI 2024**, King Abdullah University of Science and Technology (KAUST).
- Third Prize, 2015 National University Students Computer Design Competition.
- First Prize, 2014 Youth Science Popularization Innovation Competition.
- Second Prize, 2011 Chinese Physics Olympiad (provincial level).